SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1522

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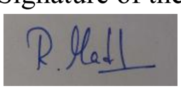
COURSE OUTCOME COVERED
CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 5

Code of Conduct
I Madhan R (192324256) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student:



Assessment Tasks
Short Answer Test
1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of	20%	Covers all	Covers most	Covers only basic	Very limited

Concepts		expected sub-points completely	expected points	points	coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Define Cloud Computing and Explain Any Four Characteristics That Make It Different from Traditional On-Premise Computing

Cloud Computing:
Cloud computing is the delivery of computing services such as servers, storage, databases, networking, and software over the Internet. It allows users to access resources on demand without owning or maintaining physical hardware.

Characteristics:

1. **On-Demand Self-Service:** Users can access computing resources whenever needed without human assistance.
2. **Broad Network Access:** Cloud services can be accessed from anywhere using devices connected to the Internet.
3. **Rapid Elasticity:** Resources can be quickly increased or decreased based on user requirements.
4. **Measured Service:** Users pay only for the resources they use (pay-as-you-go model).

Conclusion:
Cloud computing is more flexible, scalable, and cost-effective than traditional on-premise computing.

2. Trace the Evolution of Cloud Computing from Hardware Evolution to Internet Software Evolution, and Show How These Changes Enabled Modern Cloud Platforms

Cloud computing evolved through several stages of technological development:

1. **Hardware Evolution:** Early computers were large, expensive mainframes used by a single organization. Over time, powerful and affordable servers were developed.
2. **Networking Evolution:** The growth of computer networks enabled multiple systems to communicate and share resources efficiently.

3. **Internet Evolution:** The widespread availability of the Internet allowed users to access applications and data from anywhere in the world.

4. **Software Evolution:** Technologies such as virtualization, web applications, and distributed computing enabled software and services to be delivered over the Internet.

Conclusion:
These advancements in hardware, networking, the Internet, and software led to the development of modern cloud platforms, which provide scalable, reliable, and on-demand computing services such as Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).

3. Explain the Role of Server Virtualization in Cloud Computing. Differentiate Between Virtualization as an Enabling Technology and Cloud Computing as a Service Model.

Role of Server Virtualization in Cloud Computing:

Server virtualization is the process of creating multiple virtual servers on a single physical server using a hypervisor. Each virtual server works independently with its own operating system and applications. It improves resource utilization, reduces hardware costs, and makes cloud services scalable and efficient.

Difference Between Virtualization and Cloud Computing:

- Virtualization is a technology that creates multiple virtual machines on a single physical server.
- Cloud Computing is a service model that delivers computing resources over the Internet.
- Virtualization focuses on efficient hardware utilization, whereas cloud computing focuses on providing on-demand services to users.

Conclusion:
Virtualization is the foundation of cloud computing, while cloud computing uses virtualization to deliver services over the Internet.

4. Compare SOAP-Based and REST-Based Web Services in Terms of Communication Style, Data Exchange, and Suitability for Cloud Applications.

SOAP-Based Web Services:

- Uses a strict communication protocol.
- Exchanges data mainly in XML format.
- More secure and suitable for enterprise applications.

REST-Based Web Services:

- Uses standard HTTP methods such as GET, POST, PUT, and DELETE.
- Supports JSON, XML, and other formats, with JSON being the most common.
- Lightweight, faster, and ideal for cloud and mobile applications.

Conclusion:
SOAP is preferred for secure enterprise systems, while REST is widely used in cloud applications because it is simple, fast, and scalable.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with One Suitable Real-World Example for Each Model.

IaaS (Infrastructure as a Service):
Provides virtual servers, storage, and networking over the Internet.
Example: Amazon EC2 (AWS).

PaaS (Platform as a Service):
Provides a platform for developing, testing, and deploying applications.
Example: Google App Engine.

SaaS (Software as a Service):
Provides ready-to-use software that can be accessed through a web browser.
Example: Gmail.

XaaS (Anything as a Service):
Refers to any IT service delivered over the cloud, such as storage, databases, or security services.
Example: Microsoft Azure.

Conclusion:
IaaS provides infrastructure, PaaS provides a development platform, SaaS provides complete software, and XaaS is a broad term that includes all cloud-based services.